

Molecular design tools are free now

Anonymous

Preprint — August 2026

Abstract

Molecular design software no longer has to begin with procurement, a remote account, or a specialist workstation installation. Modern browsers can parse and render molecular structures, preserve chemical topology during interactive editing, and execute accelerated numerical kernels without sending a structure to a server. We describe MARCO, a browser-native molecular workspace that accepts XYZ and SMILES input, supports topology-aware fragment construction, and runs local molecular-mechanics calculations through JavaScript, WebGPU, and OpenMM 8.2 compiled to WebAssembly. The system is a concrete case study for a broader claim: a useful baseline of molecular design is now free at the points of access, installation, and data custody. “Free” does not mean that every relevant method is open, equally available on all hardware, or suitable for publication-quality prediction. The current generic force field and small-molecule GPU path are intended for interactive exploration. We therefore pair the access claim with an explicit validation program covering chemical invariants, cross-backend numerical agreement, performance, and browser portability. This framing separates what is already broadly available from the scientific and infrastructural work still required to make molecular computation genuinely universal.

Keywords: molecular design; scientific software; WebAssembly; WebGPU; molecular mechanics; browser computing

1 Introduction

For much of the history of computational chemistry, using a molecular design tool meant acquiring a license, installing a substantial native software stack, learning a remote computing environment, or some combination of the three. These requirements were not merely inconveniences. They influenced who could inspect a structure, test an idea, reproduce a calculation, and teach with the same tools used in research.

That boundary has shifted. WebAssembly brings portable compiled code to the browser [1], while WebGPU exposes local accelerators [2]. Established scientific engines can also be reused in new environments; OpenMM is one example [3]. Browser molecular graphics are already well established [4]. The newer opportunity is to place editing and calculation beside visualization without adding a remote service boundary.

This paper advances a deliberately provocative thesis: molecular design tools are free now. The useful version of that thesis is narrower than the title and more testable. A person with a contemporary browser can now access a meaningful baseline of molecular construction, visualization, and local computation without paying at the point of use, installing a native scientific stack, or surrendering molecular data to a hosted calculation service.

We examine that claim through MARCO, a browser-native molecular workspace. The contribution is not a claim that browser software has displaced high-accuracy quantum chemistry, production molecular dynamics, or expert modeling workflows. Instead, it is threefold:

1. a precise account of what “free” can mean for scientific software;
2. a reference architecture for local, browser-based molecular design; and
3. a validation framework that makes the limits of such a system visible.

2 What does “free” mean?

Price is only one barrier to molecular computation. We distinguish four freedoms that can be evaluated independently.

Free to access. The user does not pay a fee or need an institutional entitlement before the first useful interaction.

Free from installation. The application opens from a URL and does not require the user to configure a compiler, Python environment, GPU toolkit, or native molecular-simulation stack. Downloading web assets still constitutes installation in a technical sense, but the browser makes that process automatic and disposable.

Free to keep data local. Structures and trajectories need not cross a service boundary for supported operations. This property matters for unpublished compounds, classroom use, and settings in which connectivity is intermittent or policy constrains data transfer.

Free to inspect and modify. This is the traditional software-freedom question, and it depends on licensing, source availability, and the licenses of bundled components. It must not be inferred from zero price or browser delivery.

The title is therefore a hypothesis about a practical access threshold, not a claim that all molecular design software is costless, open source, or free of operational constraints. The four-part definition also supplies a more useful evaluation target than a binary label.

3 A browser-native molecular workspace

3.1 Interaction and molecular representation

MARCO presents a three-panel workspace for entering, constructing, viewing, and calculating on molecules. It accepts XYZ coordinates and SMILES strings, supports file upload, and provides a searchable fragment builder. Initial three-dimensional geometry is seeded from local chemical templates: aromatic groups are planar, saturated carbon centers are tetrahedral, neutral amines are pyramidal, and divalent oxygen and sulfur centers are bent.

Build operations preserve explicit molecular topology. Adding an atom next to an existing atom replaces an available hydrogen and adds one bond; adding it in open space begins a separate saturated component. The editor does not regenerate bonds from visual proximity, and it rejects edits that violate its valence rules. This separation between topology and rendering is essential: a molecular editor must not silently change the object being studied merely because two projected atoms appear close together.

The viewer uses an outlined ball-and-stick representation with element-specific colors and explicit bond-order lanes. Its virtual trackball uses quaternion rotation, and a rotation-invariant three-dimensional fit prevents apparent zoom changes during manipulation.

3.2 Local calculation backends

All supported calculations execute locally in workers outside the rendering thread. The lightweight JavaScript path provides immediate geometry cleanup. A browser-native WebGPU worker evaluates energy, relaxes geometry, and runs short single-precision dynamics. An OpenMM 8.2 WebAssembly worker provides a double-precision path on the Reference platform for energies, L-BFGS minimization, and short dynamics.

Both workers use MarcoFF, a compact UFF-like generic force field inspired by the broad element coverage of the Universal Force Field [5]. MarcoFF currently supports H, B, C, N, O, F, Si, P, S, Cl, Br, and I. The WebGPU path targets small molecules of at most 256 atoms. It is intended for responsive cleanup and exploratory trajectories, not for publication-quality energetics.

Calculation results retain a compact trajectory. An energy plot and frame slider allow the user to inspect saved geometries in the same viewer, making calculation history an interactive object rather than a log file hidden behind the interface.

3.3 Machine-learning feasibility boundary

The repository also contains an experimental browser export of OpenFold, whose reference implementation is a trainable reimplement of AlphaFold2 [6]. The experiment dispatches an Evoformer block and StructureModule through ONNX Runtime Web and compares WebGPU output with an in-browser WebAssembly path. Its exported models use untrained random weights. It demonstrates an execution route, not a usable structure predictor, and is not part of the current molecular-design claim.

4 Evaluation plan

A paper about access still needs conventional evidence of correctness and performance. Table 1 defines the minimum evaluation needed to turn this design report into an empirical result.

Table 1: Planned evaluation. Numerical thresholds and the device/browser matrix should be fixed before running the final benchmark.

Dimension	Test	Report
Chemical integrity	Parse, edit, attach fragments, minimize, and replay trajectories for a curated molecule set	Valence violations, topology changes, stereochemical limitations, and failing structures
Numerical agreement	Compare energies and final coordinates across JavaScript, WebGPU, and OpenMM paths where their models overlap	Absolute and relative energy differences, coordinate RMSD, convergence, and precision sensitivity
Performance	Measure cold start, first result, minimization throughput, and trajectory throughput over molecule size	Median and tail latency, memory use, device class, and thermal conditions
Portability	Run the same versioned test suite in supported browser/operating-system combinations	Feature availability, pass rate, fallbacks, and reproducible failure logs
Access	Audit the steps and external dependencies between a new user and each capability	Fees, accounts, downloads, network requests, local-data guarantees, and licenses

The existing browser regression suite is a starting point rather than the final dataset. The benchmark should be versioned, machine-readable, and runnable independently of the prose manuscript. Figures and tables should be generated from committed result files so that the reported values can be traced back to a specific application revision, browser version, and hardware configuration.

5 Limits and failure modes

Browser delivery removes some barriers and introduces others. WebGPU availability and performance vary by browser, operating system, adapter, and driver. Mobile thermal limits and memory pressure can make a nominally supported calculation impractical. Initial application assets must still be transferred and cached, and a URL is only durable if its hosting and dependency supply chain are maintained.

Scientific limits are more important. A compact generic force field cannot supply the accuracy, chemical perception, sampling depth, or uncertainty estimates required for many research conclusions. The present system covers a restricted element set and small molecules, and its interactive trajectories do not substitute for a validated simulation protocol. SMILES-to-three-dimensional initialization is useful for interaction but does not establish protonation state, tautomeric state, stereochemistry, or a physically representative conformational ensemble in every case.

Finally, privacy is an architectural property that must be verified, not merely asserted. Local calculation can coexist with analytics, remote fonts, crash reporting, or other network requests. A release supporting the data-custody claim should include a network audit, content-security policy, dependency inventory, and an offline test.

6 Discussion

The browser is becoming a credible lowest common denominator for scientific interaction. Its importance is not that it wins every benchmark against a native application. It is that one executable document can combine explanation, direct manipulation, and computation on machines that people already possess.

This changes the useful comparison. The question is no longer whether a browser can reproduce the largest or most accurate molecular workflow. It is which useful workflows can cross the zero-install, local-data threshold, how their limitations are communicated, and when a user should graduate to a specialized engine. A trustworthy tool should make that boundary obvious in the interface and in its exported results.

The word "free" also creates an obligation. If an application is free only while a particular host, browser feature, or volunteer maintainer persists, access remains fragile. Reproducible builds, permissive component licenses, archival releases, offline operation, and export to standard molecular formats are therefore part of the scientific design, not administrative details.

7 Conclusion

A useful subset of molecular design can now begin in an ordinary browser, run on local hardware, and preserve local custody of molecular structures. MARCO demonstrates this threshold for molecular construction, visualization, and exploratory mechanics while exposing clear limits in force-field fidelity, system size, and platform coverage.

The strongest form of the paper’s claim will not come from rhetoric. It will come from a public artifact, a cross-platform validation set, transparent failure reporting, and a durable licensing and

archival story. Those are the next steps required to turn “molecular design tools are free now” from a provocative title into a reproducible scientific statement.

Code and data availability

The implementation and test suite accompany this manuscript in the MARCO repository. A versioned public archive, software license, and benchmark dataset are required before public release.

References

- [1] Andreas Haas, Andreas Rossberg, Derek L. Schuff, Ben L. Titzer, Michael Holman, Dan Gohman, Luke Wagner, Alon Zakai, and JF Bastien. Bringing the web up to speed with WebAssembly. In *Proceedings of the 38th ACM SIGPLAN Conference on Programming Language Design and Implementation*, pages 185–200, 2017. doi: 10.1145/3062341.3062363.
- [2] W3C GPU for the Web Working Group. WebGPU. <https://www.w3.org/TR/webgpu/>, 2025. Accessed 2026-08-16.
- [3] Peter Eastman, Jason Swails, John D. Chodera, Robert T. McGibbon, Yutong Zhao, Kyle A. Beauchamp, Lee-Ping Wang, Andrew C. Simmonett, Matthew P. Harrigan, Chaya D. Stern, Rafal P. Wiewiora, Bernard R. Brooks, and Vijay S. Pande. OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology*, 13(7): e1005659, 2017. doi: 10.1371/journal.pcbi.1005659.
- [4] Alexander S. Rose and Peter W. Hildebrand. NGL Viewer: A web application for molecular visualization. *Nucleic Acids Research*, 43(W1):W576–W579, 2015. doi: 10.1093/nar/gkv402.
- [5] Anthony K. Rappé, Colleen J. Casewit, K. S. Colwell, William A. Goddard, and W. M. Skiff. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *Journal of the American Chemical Society*, 114(25):10024–10035, 1992. doi: 10.1021/ja00051a040.
- [6] Gustaf Ahdriz, Nazim Bouatta, Christina Floristean, Sachin Kadyan, Qinghui Xia, William Gerecke, Timothy J. O’Donnell, Daniel Berenberg, Ian Fisk, Niccolò Zanichelli, Bo Zhang, et al. OpenFold: Retraining AlphaFold2 yields new insights into its learning mechanisms and capacity for generalization. *Nature Methods*, 21(8):1514–1524, 2024. doi: 10.1038/s41592-024-02272-z.